1. Two loud speakers emit 500 Hz sound waves with an amplitude of 0.1 mm. Speaker 2 is 1.00 m behind speaker 1, and the initial phase difference between the speakers is 90°. What is the amplitude of the sound wave at a point 2.00 m in front of speaker 1? Assume that the speed of sound in air is 343 m/s.

2. Two loud speakers emit 686 Hz sound waves along the x-axis. Assume the speed of sound is 343 m/s. If the speakers are identical, what is the smallest distance between the speakers for which the interference of the sound wave is maximum destructive?

3. Noise-canceling headphones are an application of destructive interference. Each side of the headphones uses a microphone to pick up noise, delays it slightly, then rebroadcasts the noise next to your ear where it can interfere with the incoming sound wave of the noise. Suppose you are sitting 1.8 m away from an annoying 110 Hz buzzing sound. What is the minimum headphone delay in ms, that will cancel this noise?

4. Two in-phase loud speakers emit identical 1000 Hz sound waves on the x-axis. Assume the speed of sound is 340 m/s. What distance should one speaker be placed behind the other for the sound to have an amplitude of 1.5 times that of each speaker alone?

5. An interferometer is a device that uses interferes multiple paths of light in such a way to create constructive or destructive interference. Suppose that an incoming beam of radio waves with a wavelength of 15.1 m from a distant star is split in two. One beam travels 20.1 m before hitting the detector. What is the shortest distance beam 2 must trave,l that is longer than the first beam to produce an intensity maximum on the detector?

6. Two loud speakers emit sound with a wavelength of 6.1 m. The initial phase of the speaker at x = 0, leads the other speaker by 0.4 rad. Define a function,

```
Phi[x_,t_,p0] := kx -w t +p0
```

then use Nsolve[] to quickly find the locations of the first three minima and maxima.